iMak AI Lab at ArGuard Shared Tasks: Heterogeneous LoRA Ensembles of Vision-Language Models for Arabic Hateful Meme Detection

Artiom Lipinski, Ilya Makarov

iMak AI Lab, AXXX
Moscow, Russia
Correspondence: artiom0706@gmail.com

Abstract

We describe our submissions to Task A of the ArGuard 2026 shared task on harmful-content detection in Arabic memes: binary hateful-meme detection (Subtask A1) and fine-grained multilabel categorization (Subtask A2). Our A1 system is a probability-averaging ensemble of 22 fine-tuned models: 18 Qwen3-VL vision–language members (8B and 32B) trained with LoRA and scored via first-token logits, and 4 Arabic text-encoder members (MARBERTv2, AraBERTv02-twitter), combined at a fixed 0.85/0.15 modality mix. The system ranked first on Subtask A1 with 0.8234 macro-F_1, 2.2 points above our best single member. For A2, six LoRA-tuned VLMs with sigmoid multilabel heads, an A1-consistency mask, and flat 0.5 thresholds scored 0.3900 macro-F_1 and ranked second. Post-hoc analysis on the released gold labels shows that our single global threshold survived an unannounced 8-point drop in the share of hateful memes in the test set, and that most remaining errors in both subtasks separate benign sarcasm or humor from mockery aimed at a protected target.

1 Introduction

Hateful memes combine visual context, overlaid text, and cultural references, and are harder to classify than hateful text alone. The ArGuard 2026 shared task (Alam et al., 2026) covers this problem for Arabic with a fine-grained multimodal benchmark (Kmainasi et al., 2026). Task A provides Arabic memes (image plus OCR-extracted text) with a binary label (Subtask A1: *Hateful* vs. *Not Hateful*) and a 10-class multilabel fine-grained annotation (Subtask A2), evaluated by macro-F_1.

In the predecessor MAHED 2025 task (Zaghouani et al., 2025), fine-tuned models beat zero-shot prompting and off-the-shelf safety classifiers by wide margins. We therefore built our system from three components: (i) parameter-efficient fine-tuning of vision–language models — LoRA-tuned Qwen3-VL scored by the softmax over the first generated token; (ii) probability averaging across model sizes, seeds, auxiliary-label variants, training rounds, and modalities; (iii) a single global decision threshold, recalibrated once when the organizers released the development-test labels, plus a split-half stability test that rejected per-class threshold tuning for A2.

Our A1 submission ranked first among participating teams with 0.8234 macro-F_1, and our A2 submission ranked second with 0.3900.[1] With the gold test labels released after the evaluation phase, we report ablations and an error analysis, including several negative results.

2 Data and Preprocessing

We used exclusively the official `QCRI/ArGuard-Task1` data: 3,500 training, 500 development (`dev`), and 500 development-test (`dev_test`) memes with labels, and the 500-meme blind test set. After the development phase closed, the organizers released the `dev_test` gold labels and allowed their use as training data; we used them for threshold recalibration and one additional training round (§4). We used no external data and no augmentation.

Preprocessing was minimal: images were downscaled to a maximum side of 768 px, and OCR text was used as provided. The hateful prior is 37.8% in every labelled split but 29.6% in the blind test set; §5 discusses the effect of this shift.

3 System Architecture and Training

Figure 1 shows the full pipeline; the components follow below.

[1] The full official leaderboard is reported in the task overview (Alam et al., 2026).

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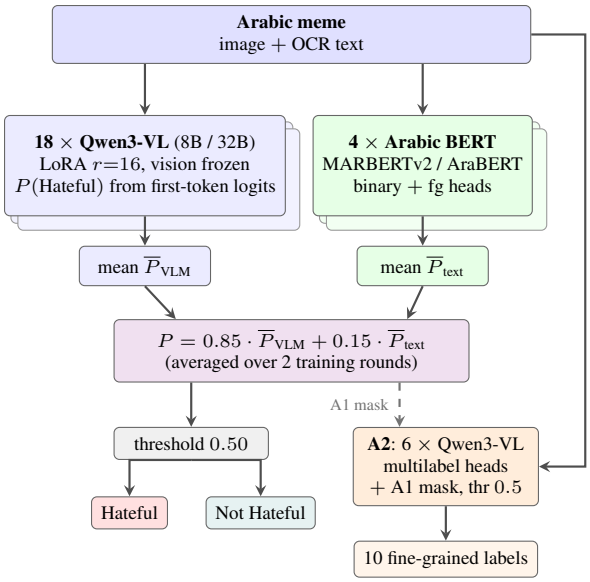


Figure 1: System overview. Each meme is scored by 18 LoRA-tuned Qwen3-VL members and 4 Arabic text-encoder members; probabilities are averaged per modality, mixed 0.85/0.15, averaged over two training rounds, and thresholded at 0.50 (Subtask A1). The six Subtask A2 members score the same meme with multilabel heads; the A1 ensemble probability only gates their outputs through a soft consistency mask.

3.1 Subtask A1

VLM members. Each meme is presented to Qwen3-VL-Instruct (Bai et al., 2025) (8B or 32B) as a chat turn containing the image and OCR text, asking whether the meme is hateful. $P(\text{Hateful})$ is the softmax over the first generated token's logits restricted to the two answer tokens. Members are fine-tuned with LoRA (Hu et al., 2022) (r=16, α=32, dropout 0.05) on all LLM linear projections with the vision tower frozen; lr 10^{-4}, cosine schedule, warmup 5%, 2 epochs, bf16, effective batch 16 (32B: micro-batch 2 × grad. accum. 8 on two A100s; 8B: 4×4 on one), gradient checkpointing. Half of the members append the A2 fine-grained labels to the training target as an auxiliary multitask signal ("fg" variants); members further differ by size (five 32B, four 8B per round) and random seed.

Text members. MARBERTv2 (Abdul-Mageed et al., 2021) and AraBERTv02-twitter (Antoun et al., 2020) are fine-tuned on OCR text with a binary head plus an auxiliary 10-class sigmoid fine-grained head (auxiliary weight 0.5); lr 2×10^{-5}, 4 epochs, batch 32.

A1 configuration (thr. 0.50)	dev_test	test
Best single member (8B, r2)	—	0.8017
Round 1 ensemble (train)	0.7645*	0.8194
Round 2 ensemble (train+dev)	0.7639	0.8254
Round 3 ensemble (+dev_test)	in-sample	**0.8268**
Rounds 1+2 (22 mem.)†	0.7701	<u>0.8234</u>
Rounds 1+2+3 (33 members)†	in-sample	0.8202
Rounds 1+2, threshold 0.45†		0.8067
Rounds 1+2, threshold 0.55†		0.8130
Rounds 1+2, oracle threshold (0.49)		0.8259

Table 1: Subtask A1 macro-F_1. Underline = official first-place submission; bold = best post-hoc configuration. † submitted during the evaluation window and scored by the live leaderboard; every other test figure in this paper is post-hoc. *`dev_test` scores computed after gold release; round 3 trains on `dev_test`, so only its test score is meaningful.

Ensembling and decision rule. Within a training round, probabilities are combined as $0.85 \cdot \overline{P}_{\text{VLM}} + 0.15 \cdot \overline{P}_{\text{text}}$ (weights selected on `dev` early in the development phase). The final submission averages two rounds, one with members trained on `train` only and one on `train+dev`, for 22 models total, thresholded at 0.50. The threshold was originally tuned on `dev` (0.57); after the `dev_test` labels were released, we recalibrated it to 0.50, which gained roughly one macro-F_1 point.

3.2 Subtask A2

Six Qwen3-VL members (four 8B, two 32B, seeds varied) share the A1 LoRA recipe plus a 10-dimensional sigmoid multilabel head trained with BCE and per-class positive weights capped at 20. Probabilities are averaged with equal weights. A soft *A1-consistency mask* halves hateful-subtype probabilities when the A1 ensemble is confident the meme is not hateful ($P \leq 0.2$) and non-hateful-subtype probabilities when it is confident the meme is hateful ($P \geq 0.8$). All classes use a flat 0.5 threshold. Per-class coordinate-ascent tuning looked attractive in-sample (0.431 on `dev_test`) but lost to flat thresholds in split-half validation (0.375 ± 0.022 vs. 0.380 ± 0.004 held-out), so we rejected it.

Infrastructure. PyTorch, HuggingFace `transformers` and `peft`, on 4×A100-80GB; one full 11-member training round takes about 9 hours.

4 Experimental Results

Submitted and post-hoc numbers. We keep two regimes apart. Before the deadline we could see only `dev`, `dev_test` and the live leaderboard, and

Modality (rounds 1+2)	thr. 0.5	oracle thr.
VLM-only (18 members)	0.8090	0.8174
Text-only (4 members)	0.7316	0.7838
0.85/0.15 mix (submitted)	**0.8234**	0.8259

Table 2: Modality ablation on test (macro-F_1). The 0.85/0.15 mix chosen on dev is also the post-hoc optimum (0.80 and 0.90 score 0.8214/0.8208).

Configuration	Mem.	GPU-h	test
1 text member	1	0.1	0.7333
1 VLM member (32B)	1	6.1	0.7913
1 VLM member (8B)	1	0.9	0.8017
4×8B + 2 text	6	3.8	0.8044
1 round (9 VLM, 2 text)	11	34.3	**0.8254**
2 rounds (submitted)†	22	68.6	0.8234
3 rounds†	33	102.8	0.8202

Table 3: Cost–accuracy trade-off (macro-F_1 at threshold 0.50). GPU-hours are measured A100-80GB training time: 6.1 h per 32B member, 0.9 h per 8B member, 0.05 h per text encoder. Inference costs one forward pass per member per meme.

we submitted five scored files, marked † in Tables 1 and 3. Every other test figure below was computed after the organizers released the gold labels, so none of it could have informed a submission.

A1. Table 1 summarizes the official result and post-hoc variants. The submitted 22-member ensemble scored 0.8234 (accuracy 0.846) and ranked first. Ensembling adds 2.2 points over the best single member, and probability averaging beats majority voting (0.8151). The text members are poorly calibrated (oracle thresholds 0.76–0.86) but still add 1.4 points at 15% weight (Table 2). Threshold probes at 0.45 and 0.55, submitted during the evaluation window, confirmed 0.50 as the peak.

How large is a meaningful difference? A 10,000-sample bootstrap over the 500 test memes puts the 95% interval of the submitted score at $[0.786, 0.858]$, with a standard deviation of 0.018. A paired bootstrap is tighter but still permissive: rounds 1+2, round 3 alone and rounds 1+2+3 span 0.8202–0.8268, and every pairwise interval contains zero. Post hoc, the round trained on all 4,500 labelled memes is the best single configuration (0.8268) even though it lowers the 33-member blend, but this test set cannot establish that ordering, and at submission time no held-out data remained to suggest it either (see Limitations). We therefore read the 0.003 between our submission and the best post-hoc variant as noise, and we treat differences below roughly 0.02 in this paper as undecided.

Cost–accuracy. Table 3 prices each configuration in measured training hours. The second training round bought nothing: one round of 11 members reaches 0.8254 post hoc, and a single-round file we did submit (threshold 0.505) scored 0.8228, within 0.001 of the 22-member submission that cost twice as much. Dropping the 32B members is the larger saving. Individually they are no better than the 8B ones (0.7913 vs. 0.8017) at 6.6 times the training cost, and they earn their place only through ensemble diversity, lifting four 8B members from 0.7991 to 0.8176. Four 8B members plus two text encoders cost 3.8 GPU-hours, 6% of the submitted system, and score 0.8044. All of these sit inside the bootstrap interval of the submission, though the cheapest ones sit at its edge. For deployment we would ship the single round, or the 8B-only ensemble under a tight compute budget; distilling the ensemble into one 8B student is the obvious next step and we did not attempt it.

A2. The submitted system scored 0.3900 macro-F_1 (micro-F_1 0.536, subset accuracy 0.144) and ranked second. The aggregate hides a split by class frequency. Over the six classes with more than 25 test instances the system reaches 0.535 (Humor 0.69, Mocking 0.66, Sarcasm 0.62, Incitement 0.45, Contempt 0.41, Other 0.38); over the four rare classes it reaches 0.173 (Dehumanization, 21 instances; Inferiority, 23; Slurs, 16; Exclusion, 3). Two measurements separate what our modelling failed to do from what the label supports allow. Oracle per-class thresholds lift the macro score to 0.4986, and 72% of that headroom comes from the four rare classes, where no threshold chosen without gold labels recovers the loss. More directly, Exclusion has three test instances: predicting a single one correctly with no false positive raises its class F_1 from 0 to 0.5 and the macro score by 0.05, which exceeds the bootstrap standard deviation of the whole metric (0.016; 95% CI $[0.360, 0.420]$). At these supports the macro average reports luck on a handful of memes as much as model quality, and the rare classes need more annotated examples rather than a better decision rule. Our modelling choices held up post hoc: removing the A1 mask costs 0.002, and the rejected `dev_test`-tuned thresholds would have scored 0.3899, so the split-half decision cost nothing.

5 Error Analysis

Prior shift and the decision threshold. The blind test set is 29.6% hateful versus 37.8% in all other splits. Our fixed threshold of 0.50 (predicting 34.6% positive) was still near-optimal: the oracle threshold of 0.49 gains only 0.0025. The F_1 curve is asymmetric, and raising the threshold to 0.60 costs 2.2 points, so a slightly generous operating point carried little risk under the shifted prior.

Error anatomy. The submission makes 77 errors: 51 false positives and 26 false negatives. False positives concentrate on non-hateful memes annotated as *Sarcasm* (33/51) and *Humor* (22/51). False negatives concentrate on *Mocking* (17/26; recall 0.83, the lowest hateful subtype, where all others reach 0.88–1.00). Both patterns involve the same distinction, separating benign sarcasm from mockery aimed at protected targets, and the same labels top the A2 confusion. Only 28 of the 77 errors have ensemble probability in $[0.35, 0.65]$; most errors are confident, and better calibration alone would not fix them. Twenty-six memes (5.2%) are misclassified by all 22 members. The ensemble corrected 8 memes against the member majority and broke 1.

Dev–test gap. All configurations score about 5 points higher on test than on `dev_test` (e.g. 0.7701 vs. 0.8234) despite the lower hateful prior. Our local `dev_test` evaluation reproduced the official development-phase scores exactly, so the gap is a difficulty difference between the two sets and our estimates were accurate.

6 Conclusion

A probability-averaged ensemble of LoRA-tuned Qwen3-VL models with a small Arabic-text-encoder component won ArGuard 2026 Subtask A1 with 0.8234 macro-F_1, and its fixed decision threshold remained near-optimal under an unannounced 8-point drop in the hateful prior. A single global threshold, recalibrated once on late-released labels, and flat A2 thresholds supported by a stability test both beat more aggressive tuning in our experiments. Pricing the ensemble afterwards is sobering: half of its 68.6 GPU-hours changed nothing, and a six-member 8B-only system holds 0.804 at 6% of the cost, so distillation is the natural next step. Future work could also model the target of sarcasm directly (sarcasm about whom?) and collect more examples of the rare fine-grained classes, which no decision rule recovers at supports of 3–23.

Limitations

Our final-file selection was made without clean held-out data: once `dev_test` labels entered training and calibration, every labelled sample was in-sample, and the post-hoc best configuration (round 3 alone, 0.8268) was indistinguishable from slightly worse blends at submission time. The cost was small (-0.003), and the bootstrap analysis in §4 shows that this difference was never measurable on 500 memes, so the missing holdout cost us less than it might have. Teams in similar phases should still reserve an internal holdout before consuming late-released labels. Our threshold probes relied on the live-scored evaluation leaderboard; if organizers hide scores in future editions, this part of our procedure becomes unavailable, although the probes only confirmed a threshold we had already set on `dev_test`. The system is compute-heavy: 68.6 A100-hours of training and 22 forward passes per meme at inference. Table 3 shows that half of that compute was unnecessary and that 6% of it stays within two points of the submitted score; we did not attempt distillation, which is the obvious way to keep the accuracy at a fraction of the cost. All conclusions come from a single 500-meme test set, where the 95% bootstrap interval of a macro-F_1 score spans roughly ± 0.036, so differences below about 0.02 should be read as undecided. Three configurations did not help: retraining on all 4,500 labelled memes improved members individually but lowered the blends we could evaluate; adding the A2 heads as extra binary A1 members left scores unchanged; and per-class A2 threshold tuning overfits at the available support sizes.

Ethical Considerations

This work uses the ArGuard dataset of hateful Arabic memes under its CC BY-NC 4.0 research licence; the content is offensive by design and was handled only for classification research, with no redistribution. Hateful-meme detectors carry dual-use risks: false positives can suppress legitimate satire (our system's most common error is over-flagging sarcasm and humor), and false negatives leave targets exposed. Deployment should keep humans in the loop and report per-subtype error rates alongside aggregate scores. The dataset contains no personally identifying annotations, and we

made no attempt to identify individuals in images. Models fine-tuned to score hatefulness could in principle be inverted to select maximally hateful content; we release no model weights.

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